DIGITALSOMA

v3.0.0

A Dynamic Physiological Digital Twin Engine for Living Animals

Technical Documentation

For researchers, veterinarians, animal scientists, and practitioners in animal health, precision livestock farming (PLF), and veterinary informatics

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Install `pip install digitalsoma`

1. What is DigitalSoma?

Imagine a pig, a horse, a cow, or a dog. That animal has a heartbeat, a body temperature, a stress response, a metabolic rate, dozens of biological signals happening simultaneously, all interconnected. Wearable sensors can now measure many of those signals continuously. But a sensor that records a number is not the same as understanding what that number means for that specific animal, at that specific moment, in that specific environmental and physiological context.

DigitalSoma is the computational counterpart of that animal. It is a software object that lives alongside the real animal and mirrors its physiology in real time. Every time a sensor sends a reading, heart rate, core temperature, cortisol, acceleration, DigitalSoma receives it, routes it through a chain of registered computational solvers, and produces a complete, internally consistent picture of the animal's current state.

The name comes from the Greek *soma* (body) and the word *digital*. A DigitalSoma is a digital body, not a static record of measurements, but a continuously self-resolving computational object.

1.1 The problems it addresses

Problem 1: The inference gap. A sensor tells you a number. It does not tell you what that number means. A heart rate of 168 bpm in a horse at full gallop is completely normal. The same number in a resting pig at 34°C ambient temperature is a clinical crisis. DigitalSoma knows the difference because it carries the species identity, the physiological baselines, the environmental context, and the computational models needed to interpret every reading.

Problem 2: Physiological fragmentation. Cardiovascular, metabolic, thermoregulatory, and stress-response signals are routinely monitored by separate tools that do not communicate. But these systems are tightly coupled in living biology, a cortisol rise amplifies the stress index, which interacts with thermal state, which changes cardiovascular demand. DigitalSoma runs all registered solvers as one connected chain, so every output reflects the full picture.

Problem 3: The gap between monitoring and reporting. Translating a clinical observation into a structured adverse event report that a regulatory authority can act on requires standardised clinical terminology, coded identifiers, and specific document formats. DigitalSoma closes this gap automatically, producing output compliant with EMA VeDDRA, HL7 FHIR R4, LOINC, and SNOMED CT.

1.2 What DigitalSoma is not

- It is not a database; it does not simply store sensor readings.
- It is not a monitoring dashboard; it actively resolves physiological meaning from incoming data.
- It is not species-specific; the same framework covers six animal species out of the box; new species can be added through the Animal Type Registry without modifying the source code.
- It is not a black box, every solver, threshold, and standard code is documented and auditable.
- It is not cloud-dependent; it runs entirely on local hardware with zero internet connection required at runtime.

2. Key Concepts

2.1 The digital twin concept

A digital twin is a computational model that mirrors a real physical entity in real time. The concept originated in aerospace engineering and was formally defined by Grieves (2014) as a virtual, digital equivalent of a physical product that shares information states across the physical and virtual domains. Tao et al. (2019) extended this framework to encompass data-driven manufacturing; DigitalSoma applies the same principle to living biology.

The real entity is the animal. The twin is the DigitalSoma software object. Every time sensors transmit data, the twin updates its internal state. The twin can then be queried, exported, or connected to clinical and regulatory pipelines.

2.2 A dynamic data object

Most software works with static data, a spreadsheet row, a database record, a configuration file. The data sits still and processes act on it from the outside. A dynamic data object is different. It resolves its own state continuously. Every time new information arrives, it recomputes all dependent quantities automatically, in the correct dependency order, and returns a fully consistent state.

DigitalSoma is a dynamic data object. Push a heart rate reading in, and the object recalculates cardiac output, adjusts the stress index, re-evaluates all VeDDRA thresholds, and updates its time-series log, all in a single synchronous call.

2.3 Ontology and semantic interoperability

An ontology is a controlled vocabulary with formally agreed meanings. It defines terms precisely and links them to globally unique identifiers so that different software systems, institutions, and countries unambiguously mean the same thing when they use the same term (Smith et al., 2007). Without ontology, every system integration requires a custom translation layer. With ontology, DigitalSoma output is machine-readable by any standards-compliant system, a veterinary EHR, a regulatory database, or a FHIR server, without any custom mapping.

3. Architecture — Five Layers

Layer	What it does
Input sources	Raw sensor data enters: wearables, implants, lab assays, remote sensing, manual entry. Fourteen-unit conversions applied automatically (°F→°C, mg/dL→mmol/L, lb→kg, etc.).
Ontology & Normalisation	Every incoming property name resolved to a canonical key. "HR", "heart_rate", "HeartRate", "pulse" all become heart_rate_bpm. Properties

	linked to Uberon, SNOMED CT, VeDDRA, NCBITaxon, LOINC, and UCUM URIs.
DigitalSoma Core	The living twin object. Three sub-layers: Structural (species identity — set once), Dynamic (live state, time-series log, threshold alarms), Functional (registered solver chain runs on every update).
LLM Agentic Interface	Eleven OpenAI-compatible tool schemas allow any language model to query and control the twin in natural language.
Outputs	State snapshot, time-series export, JSON-LD, VeDDRA adverse event report, FHIR R4 Bundle, LLM narrative.

3.1 The three core sub-layers

Structural Laye: species identity

Set once at construction via `build_soma()` and never modified at runtime. Contains the species name (taxa), NCBITaxon identifier, body mass, species-specific normal ranges for heart rate, temperature, and respiratory rate, and the list of physiological systems registered for this species. This layer provides the biological context that makes all solver computations species-aware.

Dynamic Layer: live physiological state

Maintains the animal's current state through three integrated structures: a key-value store providing O(1) access to any current property value; an append-only time-series log that records every historical state with timestamps, queryable by property and time window; and a Threshold Event System (TES) that fires configurable callback functions whenever any canonical property crosses a defined clinical boundary.

Functional Layer: the solver chain

A Model Zoo of registered solver functions that executes as a Directed Acyclic Graph (DAG) on every `update_sync()` call. Each solver receives the current params and state dicts and returns derived quantities that are written back into the state. DigitalSoma ships with six built-in solvers as initialisation scaffolding — designed to be supplemented, overridden, or replaced by user-defined solvers implementing any computational approach: data-driven (statistical models, regression equations, machine learning) or mechanistic (compartmental models, differential equations, allometric scaling).

4. Built-In Solvers and the Extensible Solver Chain

DigitalSoma ships with six built-in solvers. These are provided as a working starting point, not as a fixed computational framework. Each can be replaced, extended, or run alongside additional user-

defined solvers. The solver chain is fully composable, any function accepting a params dict and a state dict and returning a dict of new properties can be registered.

What a solver is

Signature: `fn(params: dict, state: dict)` ➔ dict

It reads from current state, performs any computation, and returns new or updated properties.

No constraint on computational approach: allometric equations, regression models, machine

learning inference, compartmental models, differential equations, empirical lookup tables.

4.1 The six built-in solvers

Solver 1 — Cardiovascular baseline

Outputs

`cardiac_output_L_min`: volume of blood pumped per minute

`mean_arterial_pressure_mmHg`: estimated from systolic and diastolic pressure when available

Estimates cardiac output from heart rate and species-specific stroke volume using the Fick approximation (Fick, 1870; Guyton & Hall, 2015). Cardiac output equals heart rate multiplied by stroke volume, providing a computationally lightweight cardiovascular baseline without invasive instrumentation.

```
Cardiac Output (L/min) = Heart Rate (bpm) × Stroke Volume (mL) ÷ 1000
Example: HR = 75 bpm, Stroke Volume = 60 mL (porcine default)
         Cardiac Output = 75 × 60 ÷ 1000 = 4.5 L/min
```

Solver 2: Metabolic rate

Outputs

`rmr_kcal_day`: resting metabolic rate in kcal/day

`rmr_W`: resting metabolic rate in watts

Computes resting metabolic rate using Kleiber's allometric scaling law (Kleiber, 1947, 1961) with a Q10 temperature correction (Blaxter, 1989). The Q10 coefficient (default 2.0) reflects the well-established biochemical principle that metabolic rate approximately doubles for every 10°C rise in core temperature.

```
RMR (kcal/day) = 70 × body_mass_kg ^ 0.75
RMR_adjusted   = RMR × Q10 ^ ((T_core - T_base) / 10)
Example: 90 kg pig, T_core = 40.1°C, T_base = 38.8°C, Q10 = 2.0
         RMR_base = 70 × 90^0.75 = 1827 kcal/day
```

```
RMR_adjusted = 1827 × 2.0^(1.3/10) = 2004 kcal/day
```

Solver 3: Thermoregulation

Outputs

`heat_loss_W`: heat dissipated to the environment

`net_heat_flux_W`: net thermal balance

`thermal_comfort_index` (TCI): dimensionless: 0 = balanced, >0 = heat stress, <0 = cold stress

Computes a Thermal Comfort Index (TCI) by comparing metabolic heat production against heat loss to the environment using Newton's law of cooling (Newton, 1701; Mount, 1979; Gebremedhin & Wu, 2001). The species-specific thermal conductance parameter (k) scales with the square root of body mass.

```
heat_loss (W) = k × (T_core - T_ambient)
where k = 0.05 × √body_mass_kg   (species-specific thermal conductance)
TCI = (metabolic_rate_W - heat_loss_W) / metabolic_rate_W
```

Solver 4: Respiratory gas exchange

Outputs

`vo2_L_min`: oxygen consumption

`vco2_L_min`: carbon dioxide production

`minute_ventilation_L_min`: total air volume per minute

Derives oxygen consumption from metabolic power output using the caloric equivalent of oxygen, approximately 4.8 kcal per litre O_2 (Brouwer, 1965; West, 2012). Carbon dioxide production uses the respiratory quotient (default RQ = 0.85, reflecting mixed substrate combustion; McLean & Tobin, 1987). Minute ventilation is the product of respiratory rate and tidal volume (Weibel, 1984).

Solver 5: Physiological Stress Index (PSI)

Outputs

`physiological_stress_index`: composite 0-to-1 score

Computes a Physiological Stress Index (PSI) by combining three independently measurable stress signals into an equally weighted mean: heart rate deviation from species baseline, cortisol elevation above resting baseline, and absolute thermal discomfort.

```
HR_deviation  = |HR - HR_base| / HR_base
Cortisol_dev  = max(0, (cortisol - cortisol_base) / cortisol_base)
Thermal_dev   = |TCI|
```

```
PSI = (HR_deviation + Cortisol_dev + Thermal_dev) / 3

PSI > 0.70  →  triggers VeDDRA Dyspnoea flag (PT 506)
PSI > 0.40  →  elevated — monitor closely
PSI < 0.20  →  low stress
```

⚠️ Limitation and honest scope of the PSI

The Physiological Stress Index (PSI) is an **original composite formulation developed specifically for DigitalSoma**. It serves two purposes within this framework only:

- To trigger the Threshold Event System (TES) alarm when multi-system stress is detected simultaneously.
- To provide an input to the VeDDRA Dyspnoea flag (PT 506) as a proxy for generalised physiological distress in the absence of a direct respiratory distress sensor.

The individual components draw on established physiological stress indicators, heart rate deviation (von Borell et al., 2007), cortisol as an HPA axis marker (Mormède et al., 2007), and thermal discomfort (Mount, 1979), but **the composite formula, the equal weighting between components, and the 0.70 threshold have not been validated** against independent welfare assessments or clinical outcomes in any species.

The PSI must not be used as a standalone clinical diagnostic criterion or welfare certification instrument. Species-specific and context-specific validation studies are required before the PSI can support clinical decision-making. Researchers are encouraged to replace the built-in PSI solver with validated welfare indices appropriate to their species and study context.

Solver 6: VeDDRA adverse event screen (v3.0.0)

Outputs

`ae_flags`: list of VeDDRA flags: PT code, preferred term, SOC, HLT, SNOMED CT concept

`adverse_event_score`: ratio of fired flags to total species-specific rules (0–1)

The species-aware VeDDRA adverse event screen introduced in v3.0.0. Evaluates a curated set of VeDDRA Preferred Term (PT) rules for the specific animal type in the ATR, using official EMA PT codes from VeDDRA Rev.16 (EMA, 2025). All thresholds are species-relative. VeDDRA codes are official EMA Rev.16 PT-level integer codes, corrected in v3.0.0 from the incorrect 8-digit codes used in v2.x.

VeDDRA PT	PT code	Trigger (species-relative)
Hyperthermia	604	T_core > T_base + 1.5°C
Hypothermia	605	T_core < T_base − 2.0°C
Tachycardia	122	HR > HR_base × 1.5

Bradycardia	120	HR < HR_base × 0.6
Hypoxia	501	SpO_2 < 90%
Tachypnoea	515	RR > RR_base × 1.5
Bradypnoea	504	RR < RR_base × 0.5
Dyspnoea	506	PSI > 0.70
Dehydration	657	Haematocrit > species threshold
Anaemia	85	Haemoglobin < species threshold
Hypoglycaemia	904	Glucose < species threshold
Hyperglycaemia	903	Glucose > species threshold
Cyanosis	601	SpO_2 < 86% (severe)

4.2 Adding user-defined solvers

Any function conforming to the solver signature can be registered with `ds.register_method()`. It receives the full current state on every `update_sync()` call. There is no constraint on the computational method.

```
def welfare_solver(params: dict, state: dict) -> dict:
    """Example user-defined welfare index solver."""
    tci  = state.get("thermal_comfort_index", 0.0)
    psi  = state.get("physiological_stress_index", 0.0)
    spo2 = state.get("spo2_pct", 98.0)
    thermal_w = max(0.0, 1.0 - abs(tci) * 2.0)
    stress_w  = max(0.0, 1.0 - psi)
    oxygen_w  = max(0.0, min(1.0, (spo2 - 90.0) / 8.0))
    awi = 0.40*thermal_w + 0.40*stress_w + 0.20*oxygen_w
    return {"animal_welfare_index": round(awi, 4)}

ds.register_method("welfare_index", welfare_solver)
# To replace a built-in solver:
ds.unregister_method("metabolic_rate")
ds.register_method("metabolic_rate", my_mechanistic_model)
```

5. Animal Type Registry

The Animal Type Registry (ATR) is the catalogue of species templates built into DigitalSoma. When a twin is constructed with `build_soma()`, the ATR template for the specified species is loaded into the structural layer, providing species-specific baselines used by all solvers for relative threshold computation.

Registry key	Species	Key parameters
`porcine_adult`	Sus scrofa domesticus (pig)	90 kg · HR 75 bpm · T 38.8°C · cortisol 40 nmol/L
`equine_adult`	Equus caballus (horse)	500 kg · HR 36 bpm · T 37.8°C · RR 12 bpm
`bovine_adult`	Bos taurus (cattle)	600 kg · HR 60 bpm · T 38.5°C · RR 20 bpm
`ovine_adult`	Ovis aries (sheep)	70 kg · HR 75 bpm · T 39.0°C · RR 25 bpm
`canine_adult`	Canis lupus familiaris (dog)	25 kg · HR 90 bpm · T 38.5°C · RR 22 bpm
`salmonid_adult`	Salmo salar (Atlantic salmon)	4.5 kg · HR 50 bpm · ectotherm

5.1 Adding a custom species

```
from digitalsoma import register_animal_type, build_soma, SomaConfig

register_animal_type("alpaca_adult", {
    "taxa": "Vicugna pacos",    "ncbi_taxon_id": "30538",
    "body_mass_kg": 65.0,       "core_temp_normal_C": 38.0,
    "hr_normal_bpm": 70.0,      "rr_normal_bpm": 20.0,
    "systems": ["cardiovascular", "metabolic", "thermoregulation"],
})
ds = build_soma(SomaConfig(animal_type="alpaca_adult"))
```

6. Ontology Layer and International Standards

DigitalSoma implements a multi-standard semantic layer that anchors every canonical property to internationally recognised ontology URIs. This layer normalises incoming data (resolving vendor aliases to canonical keys) and annotates outgoing data (attaching globally unique identifiers in JSON-LD exports and FHIR resources).

6.1 How the standards connect: “the data flow”

The diagram below traces a sensor reading from entry through to regulatory-grade output, showing where each standard is engaged.

```
Input:   HR = 130 bpm, T = 40.6°C, SpO2 = 95.8%
```

↓ **Step 1: Ontology & Normalisation**

Uberon · SNOMED CT · UCUM · NCBITaxon · PATO

```
canonical_key("HR")     → heart_rate_bpm    [URI: CMO_0000052]
canonical_key("T")      → core_temp_C       [URI: snomed/276885007]
canonical_key("SpO2")   → spo2_pct          [URI: snomed/103228002]
```

↓ **Step 2: DigitalSoma Core (solver chain)**

NCBITaxon identifies species → ATR loads porcine_adult baselines

```
cardiac_output_L_min        = 7.8 L/min
physiological_stress_index  = 0.88  (alarm triggered)
ae_flags: [PT 604 Hyperthermia, PT 122 Tachycardia, PT 515 Tachypnoea]
```

↓ **Step 3: Output encoding**

VeDDRA Rev.16 PT codes: 604, 122, 515
SNOMED CT clinical concept codes: 386689009, 3424008, 271823003
LOINC observation codes: 8867-4, 8310-5, 59408-5
UCUM units: /min, Cel, %
HL7 FHIR R4 Bundle: Patient + Device + 21 Observations + DiagnosticReport
JSON-LD @context mapping all URIs; FAIR-compliant linked data

6.2 The six ontology namespaces[1]

Namespace	Purpose and URI
Uberon	Cross-species anatomy; anchors every property to its anatomical/physiological concept http://purl.obolibrary.org/obo/UBERON_
SNOMED CT	Clinical concept identity; `vocab.py` URIs and FHIR Observation dual coding http://snomed.info/sct
NCBITaxon	Species identity; ATR and FHIR Patient resource http://purl.obolibrary.org/obo/NCBITaxon_
UCUM	Units of measure; all FHIR Observation valueQuantity units http://unitsofmeasure.org
LOINC	Laboratory and clinical observation codes; 23 properties mapped for FHIR http://loinc.org

[1] A named container that groups related identifiers together and makes them globally unique by prefixing them with a shared address.

VeDDRA	Veterinary adverse event terminology; EMA Rev.16 PT codes https://www.ema.europa.eu/en/veterinary-regulatory/

6.3 Alias resolution

```
from digitalsoma.ontology.vocab import canonical_key
canonical_key("HR")           # → heart_rate_bpm
canonical_key("SpO2")         # → spo2_pct
canonical_key("Tb")           # → core_temp_C
canonical_key("HeartRate")    # → heart_rate_bpm
canonical_key("pulse")        # → heart_rate_bpm
canonical_key("cort")         # → cortisol_nmol_L
```

7. HL7 FHIR R4 Integration

DigitalSoma implements a native HL7 FHIR R4 I/O layer through the `digitalsoma.fhir` subpackage, using Python's standard library only; zero external dependencies.

FHIR Resource	What it contains
Patient	Animal subject. NCBITaxon species extension (e.g. NCBITaxon:9823 for pig). Animal ID and site name.
Device	The DigitalSoma twin itself. Identified by `soma_id`. Version 3.0.0. Linked to Patient.
Observation (×21)	One per mapped canonical property. Dual-coded LOINC + SNOMED CT. UCUM units. `canonical_key` extension for lossless round-trips.
DiagnosticReport	VeDDRA flags mapped to SNOMED CT + VeDDRA PT code. Status: final (no flags) or amended (flags active).

```
from digitalsoma import build_soma, SomaConfig, to_fhir_bundle,
from_fhir_bundle
ds = build_soma(SomaConfig(animal_type="porcine_adult", animal_id="pig-001"))
ds.update_sync({"heart_rate_bpm": 130, "core_temp_C": 40.6, "spo2_pct": 95.8})
bundle   = ds.to_fhir_bundle()                          # collection bundle
tx       = ds.to_fhir_bundle(bundle_type="transaction") # for POST to FHIR
server
readings = from_fhir_bundle(incoming_bundle)            # parse → update_sync()
```

8. VeDDRA Pharmacovigilance — v3.0.0

The term vocabulary is derived directly from the official EMA dataload file (EMA/CVMP/PhVWP/10418/2009 Rev.16, effective 1 October 2025), embedded in `digitalsoma/veddra/veddra_terms.py`. All codes are correct EMA Rev.16 PT-level integer codes, corrected from the incorrect 8-digit codes used in v2.x. No internet connection is required at runtime.

Species	v3.0.0 rules (up from 6 in v2.x)
`porcine_adult`	13
`equine_adult`	12
`bovine_adult`	12
`ovine_adult`	12
`canine_adult`	13
`salmonid_adult`	4 (ectotherm; no temperature rules)

Every VeDDRA flag in the `ae_flags` output carries the official EMA PT code, the Preferred Term display name, the SOC and HLT hierarchy, and the corresponding SNOMED CT concept ID. The FHIR DiagnosticReport encodes each flag with dual VeDDRA + SNOMED CT coding, making it simultaneously readable by regulatory systems (EMA EVVET3, UK VMD, FDA-CVM) and clinical decision support systems.

9. Quick Start

9.1 Installation

```
pip install digitalsoma

python -c "import digitalsoma; print(digitalsoma.__version__)"
# → 3.0.0
```

9.2 First twin: “only ten lines”

```
from digitalsoma import build_soma, SomaConfig, to_fhir_bundle

ds = build_soma(SomaConfig(
    animal_type = "porcine_adult",
    animal_id   = "pig-barn-A-01",
    site_name   = "Research Unit 3, University of Manitoba",
))
```

```
state = ds.update_sync({
    "HR":   75,       # alias → heart_rate_bpm
    "Tb":   38.8,     # alias → core_temp_C
    "SpO2": 98.5,     # alias → spo2_pct
    "RR":   20,       # alias → respiratory_rate_bpm
})

print(state["cardiac_output_L_min"])         # 4.5 L/min
print(state["thermal_comfort_index"])         # ~0.0 (balanced)
print(state["physiological_stress_index"])   # ~0.02 (low stress)
print(state["ae_flags"])                     # [] (no adverse events)

bundle = ds.to_fhir_bundle()   # HL7 FHIR R4 Bundle export
```

10. LLM Agentic Interface

Tool	What it does
`ds_describe`	Full natural-language summary of current physiological state
`ds_get_state`	Retrieve the value of a specific canonical property
`ds_update`	Ingest sensor readings and run the full solver chain
`ds_query_history`	Retrieve time-series for a property (limit, since)
`ds_list_solvers`	List all registered solvers in DAG order
`ds_to_jsonld`	Export current state as JSON-LD linked data
`ds_veddra_report`	Generate structured VeDDRA adverse event report
`ds_alarm_status`	List all TES alarms currently fired
`ds_manifest_summary`	List all registered sensors
`ds_structural_layer`	Return species identity and normal ranges
`ds_to_fhir_bundle`	Export state as HL7 FHIR R4 Bundle

11. Output Formats

Output format	Description
State snapshot	Python dict of all current canonical properties and solver outputs. JSON-serialisable.
Time-series log	Timestamped list of historical snapshots. Queryable by property and time window.
JSON-LD	W3C linked data with full ontology URIs. FAIR-compliant. Every property has a globally unique identifier.
VeDDRA report	Structured adverse event report with official EMA PT codes, SOC/HLT hierarchy, and SNOMED CT mappings.
FHIR R4 Bundle	Patient + Device + Observations (LOINC + SNOMED CT + UCUM) + DiagnosticReport. Compatible with EMA EVVET3, UK VMD, FDA-CVM.
LLM narrative	Natural language clinical summary via connected language model tool interface.

12. Standards and Data Models — Complete Reference

Standard	Layer	Status in v3.0.0
Uberon	Ontology	Full; property URI anchoring
SNOMED CT	Ontology + FHIR	Full; `vocab.py` URIs + FHIR Observation coding
NCBITaxon	Structural + FHIR	Full; ATR and FHIR Patient
UCUM	Ontology + FHIR	Full; all FHIR Observation units
LOINC	FHIR	Full; 23 properties mapped
VeDDRA Rev.16	Functional + Output	Full; species-aware, official EMA codes
HL7 FHIR R4	Output	Full; 4 resource types
JSON-LD (W3C)	Output	Full; FAIR linked data

PATO / HP / MP	Ontology	URI vocabulary only
OMOP CDM	—	Not included
openEHR	—	Not included
ICAR	—	Not included

13. Citation and Licence

13.1 How to cite

```
Youssef, A. (2026). DigitalSoma: A Dynamic Physiological Digital Twin Engine
for Living Animals (v3.0.0). Digital Agroecosystems Laboratory,
University of Manitoba & BioTwinRs Ltd., Leuven, Belgium.
https://github.com/Pierianspring/digitalsom | ORCID: 0000-0002-9986-5324
```

13.2 Licence

Released under the Creative Commons Attribution 4.0 International licence (CC BY 4.0). Free to use, share, modify, and redistribute for any purpose, including commercial, with attribution.

13.3 Contact

- GitHub: https://github.com/Pierianspring/digitalsoma
- PyPI: https://pypi.org/project/digitalsoma
- Email: info@biotwinrs.com
- ORCID: https://orcid.org/0000-0002-9986-5324

References

References are grouped by solver and topic. All built-in solvers reference the primary literature cited below. The Physiological Stress Index (Solver 5) is an original formulation by the author; its individual components draw on the cited sources, but the composite index itself has not been independently validated and must not be used as a clinical diagnostic instrument.

Digital twin concept

Grieves, M. (2014). *Digital twin: Manufacturing excellence through virtual factory replication.* White Paper. Florida Institute of Technology.

Tao, F., Sui, F., Liu, A., et al. (2019). *Digital twin-driven product design framework.* International Journal of Production Research, 57(12), 3935–3953.

Björnsson, B., et al. (2020). *Digital twins to personalize medicine.* Genome Medicine, 12(4).

Solver 1: Cardiovascular (Fick equation)

Fick, A. (1870). *Über die Messung des Blutquantums in den Herzventrikeln.* Sitzungsberichte der Physikalisch-Medizinischen Gesellschaft zu Würzburg, 2, 16–28.

Guyton, A. C., & Hall, J. E. (2015). *Textbook of Medical Physiology (13th ed.).* Elsevier Saunders.

Evans, D. L., & Rose, R. J. (1988). *Cardiovascular and respiratory responses to exercise in thoroughbred horses.* Journal of Experimental Biology, 134(1), 397–408.

Muir, W. W., & Hubbell, J. A. E. (2009). *Equine Anesthesia: Monitoring and Emergency Therapy (2nd ed.).* Saunders Elsevier.

Solver 2: Metabolic rate (Kleiber's law + Q10)

Kleiber, M. (1947). *Body size and metabolic rate.* Physiological Reviews, 27(4), 511–541.

Kleiber, M. (1961). *The Fire of Life: An Introduction to Animal Energetics.* Wiley, New York.

Schmidt-Nielsen, K. (1984). *Scaling: Why is Animal Size so Important?.* Cambridge University Press.

Blaxter, K. L. (1989). *Energy Metabolism in Animals and Man.* Cambridge University Press.

Brown, J. H., Gillooly, J. F., Allen, A. P., Savage, V. M., & West, G. B. (2004). *Toward a metabolic theory of ecology.* Ecology, 85(7), 1771–1789.

Solver 3: Thermoregulation (Newton's law of cooling)

Newton, I. (1701). *Scala graduum caloris.* Philosophical Transactions of the Royal Society, 22, 824–829.

Mount, L. E. (1979). *Adaptation to Thermal Environment: Man and His Productive Animals.* Edward Arnold, London.

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Mader, T. L., Davis, M. S., & Brown-Brandl, T. (2006). *Environmental factors influencing heat stress in feedlot cattle.* Journal of Animal Science, 84(3), 712–719.

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Solver 4: Respiratory gas exchange

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